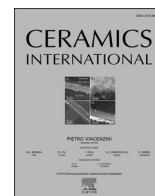




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Review article

Mechanical modeling and simulation of aerogels: A review

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ARTICLE INFO

Keywords:

Aerogels

Mechanical modeling

Simulations

Mechanical properties

ABSTRACT

Aerogels are solids with exceptional characteristics, such as ultra-low density, high surface area, high porosity, high adsorption and low-thermal conductivity. Due to these characteristics, aerogels are emerging to be a popular material among the scientific community, since their discovery in 1931. However, their applicability has remained questionable due to the poor mechanical characteristics, such as brittleness and low tensile strength. In the last three decades, due to the rapid development in the computational resources and numerical methods, a deeper understanding of physical behavior and properties of these materials have been extensively investigated. In this work, an effort is made to critically analyze and categorize the computational models and simulation results available for silica, carbon, carbon nanotubes, graphene, and cellulose aerogels. This work focused on a better understanding of how these materials were computationally modeled and simulated over the time-period and at different length-scales, wherein primary approaches, such as molecular dynamics (MD), coarse-grained, micromechanical multiscale and continuum mechanics modeling, were discussed. It also strives to give an insight into the areas where further computational studies are required, which could lead to numerous other application fields. The systematic review provides a mechanistic basis for reliable applications of aerogels.

1. Introduction

In recent years, the sol-gel field grown the interest in material scientists due to its gel formation process that has some degree of ordering and structure. The sol-gel process enables a great control over particle size and morphology of the prepared materials [1–3]. The most famous nanoporous material synthesized using the sol-gel process is aerogel. Charles Learned once made a bet with Kistler over the possibility of shrinkage-free replacement of liquid in “jellies”. To win the bet, Kistler performed different experiments, which formed the “aerogel compounds” in 1931 [4]. With silica as the solid base in the gel, Kistler was successful in the discovery of the first-ever silica aerogels. Supercritical drying of the liquid that constituted the gel was the primary factor governing the synthesis of aerogels. It needed the gel to be heated so that the entrapped liquid in the gel would exceed the critical values of pressure and temperature, which allowed the supercritical fluid in gaseous form to be separated from the material (see Fig. 1). The supercritical drying of aerogels forms aerogel with higher pore volumes and porosities, and higher surface areas compared to the freeze-drying and ambient pressure drying. At supercritical conditions, the solvent's surface tension during synthesis becomes zero, minimizing the drying

stress to form a monolithic and highly porous aerogel network. These voids formed the basis of several mechanical behavioral properties of the aerogels. One can say with certainty that the structure is the topologically optimized structure of a sol-gel.

Extensive experimental studies have been performed on silica, carbon, graphene, and cellulose aerogels over the last several decades to understand their structural, mechanical, thermal, and electrical properties under varied circumstances [5–36]. The exciting physical properties, such as high porosity, high specific surface area, low thermal conductivity, and better electrical conductivity (in case of carbon-based) of aerogels, open the way towards several exciting applications. Namely, thermal insulation, catalysis, oil-spill cleanup, sensors, interlayer dielectrics, and drug delivery. Presently, the aerogels are widely used in photocatalytic degradation of water and gaseous pollutants, photo-/electrocatalysis, hybrid supercapacitors, lithium-ion batteries, phase change materials, triboelectric nano-generators, piezoelectric sensors, and various interesting applications [37–45].

In recent years, researchers have benefited from tremendous progress in theoretical studies, numerical methods, and computer science, coupled with high-performance computing resources. Nowadays, the large scale simulations in engineering and material sciences are possible

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Received 3 August 2020; Received in revised form 16 September 2020; Accepted 18 September 2020

Available online 28 September 2020

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because of new technology. A deep understanding of the relationships between the structure and its properties is facilitated by computational advances from the nanoscale to the continuum scale. For example, molecular dynamics (MD) simulations with a model size of up to a few million atoms can now be performed to investigate the structural, thermal, mechanical, and fracture properties of aerogels [46–52]. The mechanical modeling and simulations can be used for several purposes, such as, to provide a different length-scale picture of structure and mechanics allows the prediction of properties for novel materials which are impossible to be investigated experimentally, investigating properties of conceptual (theoretical) materials, and substituting the experimental methods for obtaining the mechanical parameters.

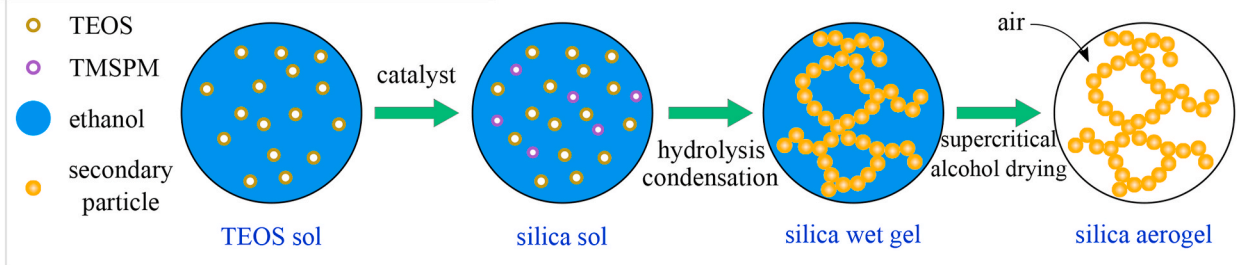
Thus, the last three decades have seen a progressive rise in the use of computational modeling and simulations in aerogel studies. This review provides a comprehensive overview of the mechanical properties of various aerogels (silica, graphene, CNT, cellulose, etc.) prepared by different drying methods. This review is organized systematically in different sections. In Section 2, the mechanical modeling and simulations of silica aerogels are described using MD simulations, coarse-grained modeling and continuum mechanics. Section 3 briefly describes the mechanical modeling of carbon and carbon nanotube aerogels, which is not very exhaustive. Moreover, the recent advances in the modeling of a newly developed material, graphene aerogel, is presented in Section 4. Furthermore, in Section 5, the available continuum models

of cellulose aerogels are discussed. Finally, the summary and future perspectives of modeling of aerogels are presented in the last section.

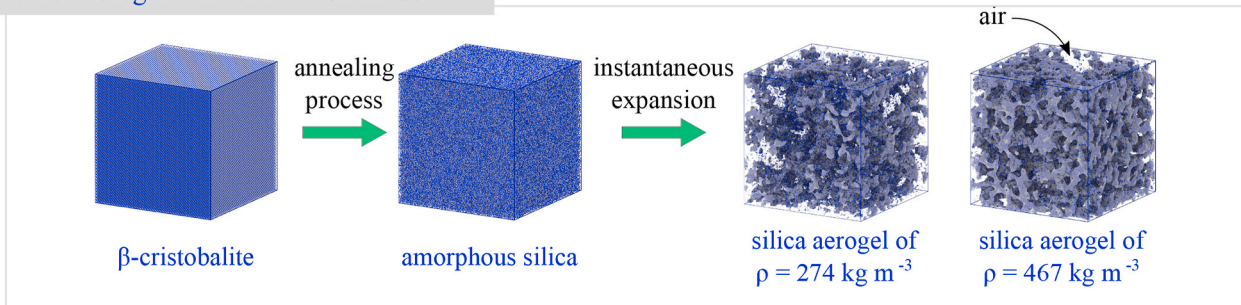
2. Silica aerogels

Silica aerogels are currently one of the most extensively researched aerogel materials. They are prepared by supercritically drying silica gels, which were also incidentally, the first inorganic aerogels to be created. Supercritical drying is possible at different temperatures. Depending on the temperature and the additional reagent used, it is possible to obtain silica aerogels with different densities and properties [53]. The crown of the “material with lowest density” was held by silica aerogel until aerographite was discovered, which was again bested by aerographene (graphene aerogel) in 2013. By virtue of their unique properties, silica aerogels are suitable for a wide variety of engineering applications. Amongst the most notable applications in recent times, silica aerogel was used to regulate the temperatures in aerospace applications [54]. On the Stardust Mission, NASA again found silica aerogels useful to trap space dust as they are highly porous. This characteristic also makes silica aerogel a competent candidate for filtration and adsorption processes. In the research of particle physics, silica aerogels are used as Cherenkov effect detectors for particle identification [55]. They are used in imaging devices, optics, and light guides as they possess a low refractive index and excellent capacity to scatter light.

Silica aerogel: experimental procedure



Silica aerogel: MD simulation model



Graphene aerogel: MD simulation model

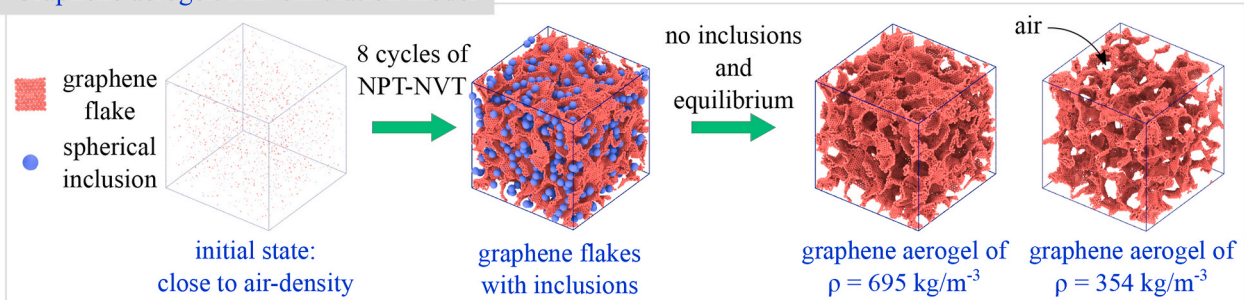


Fig. 1. Schematic representation of (a) the experimental process of silica aerogel, (b) creation of the MD model of silica aerogel, and (c) mimicking experimental procedure for the formation of an all-atom model of graphene aerogel.

Multiple experimental studies made by various researchers, like Schwertfeger et al. [17–19], A. V. Rao et al. [20,23,24,27,30–36,56–62], Fricke et al. [6–8,10–16,63], show silica aerogels of a range of densities and provide an important basis for multitude of mentioned properties. Presently, silica aerogels are very well commercialized by well-known aerogel industries, such as Aspen Aerogels, Cabot Corporation, and many more industries situated in America, Europe and Asia [64]. In addition to costs, experimental studies have an inherent limitation of not being able to cover multiple aspects at the same time with the same sample. For example, the stiffness of the aerogel is highly dependent on the solid network connectivity, which is different for the same density. Unfortunately, there are no consolidated experiments done to measure this connectivity. As a viable and effective solution to this cost-ease conundrum, Kieffer and Angell in 1988 found to aid in simulation sciences and computational models for analyzing aerogels [65]. As an integral part of studying aerogels, few researchers like Gelb and Ferreiro-Rangel have made impressive strides in mechanical modeling by developing computational methods for determination of mechanical properties of silica aerogels and analyzing their impact on fracture mechanics and applications [66–69]. Computational modeling of silica aerogels is done in four significant ways – molecular dynamics, coarse-grained models, micromechanical multiscale models, and continuum models. Approaches used by different models to simulate the unique atomic composition of the silica aerogels vary to achieve different objectives. The selection of an approach is based on the nature of the aerogel characteristic to be studied. The following sections explain in detail these approaches as perceived by different researchers and their results pertaining to the mechanical and thermal properties.

2.1. Molecular dynamics simulations

Molecular Dynamics (MD) simulation is a computational method employed for assessing the interparticle interactions in the molecular level, dependant on time. These simulations numerically solves Newton's equations of classical motion,

$$m_i \frac{\partial^2 \mathbf{r}_i}{\partial t^2} = \mathbf{F}_i, \text{ and } \mathbf{F}_i = -\nabla_{\mathbf{r}_i} U(\mathbf{r}) \quad i = 1, 2, \dots, N. \quad (1)$$

N is number of interacting atoms in the MD system. $\mathbf{r}_i$ and m_i represent the position and mass of atom i , respectively. $\mathbf{F}_i$ is the force acting on atom i .

The systems entering MD simulations are modeled as ensembles of interacting particles under specific internal and external conditions [47–52,70]. Interacting under certain conditions like cyclic loading in an explicit scheme, the interactions are governed by the force fields or the equations of motion. They are primarily used to determine the interatomic potentials of the component under observation. They provide a comprehensive understanding of the bonding, non-bonding interactions of particles with neighbors.

Force-field can be separated into bonded and non-bonded interactions:

$$U(\mathbf{r}) = U_{\text{bonded}}(\mathbf{r}) + U_{\text{non-bonded}}(\mathbf{r}). \quad (2)$$

Bonded interactions include bond-stretching ($U_{\text{stretching}}$), bending (U_{angles}), torsion (U_{torsion}), and out-of-plane improper torsion (U_{improper}). In particular,

$$\begin{aligned} U_{\text{bonded}}(\mathbf{r}) &= U_{\text{stretching}}(\mathbf{r}) + U_{\text{angles}}(\mathbf{r}) + U_{\text{torsion}}(\mathbf{r}) + U_{\text{improper}}(\mathbf{r}) \\ &= \sum_{\text{stretching } i} K_{ri}(r_i - r_0)^2 + \sum_{\text{angles } i} K_{\theta i}(\theta_i - \theta_0)^2 \\ &+ \sum_{\text{torsion } i} \frac{V_i}{2} [1 + \cos(n\varphi_i - \varphi_0)] + \sum_{\text{improper } i} K_{\xi i}(\xi_i - \xi_0)^2. \end{aligned} \quad (3)$$

Here, K_{ri} representing the spring constant, and r_0 is the bond length at equilibrium. Also, $K_{\theta i}$ is the force constant, θ_0 is the equilibrium angle, V_i is the coefficient in the Fourier series, φ_0 is the torsional angle at

equilibrium, φ_i is the actual torsional angle, n is the number of minima or maxima between 0 and 2π , $K_{\xi i}$ is the large spring constant, ξ_0 is the improper angle at equilibrium, and ξ_i represents the actual improper angle.

The non-bonded interactions encompass Lennard-Jones and Coulombic interactions, that is

$$U_{\text{non-bonded}}(\mathbf{r}) = \sum_i \sum_j \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \right\}, \quad (4)$$

where the first two terms of Equation (4) is the Lennard-Jones potential. Here, r_{ij} is the distance between atom i and j , ϵ is the depth of energy well, and $\sigma = r/2^{1/6}$ is the distance is the finite distance at which the potential is zero. The last term of Equation (4) is the Coulombic potential, wherein q_i and q_j are the charges of the atoms i and j , respectively. Moreover, ϵ_0 represents the permittivity.

Also, there are several theories put forth by researchers, such as Haile et al. [71]; Frenkel and Smit [72]; Rapaport 2004 [73] and Allen and Tildesley [74], which provide different approaches to adapting the molecular dynamics for different materials. In the various theories, several force fields are theorized to adapt this method to simulate the silica aerogel models. However, the implementation of these theories into reality and validating have been carried out by only a handful because of their computational complexity.

The recent forays in the adaptation of molecular dynamics to analyze silica aerogels are governed by the Vasishta Potential which considers the effects of 2 and 3 body interactions with the following equations [70, 75,76].

$$U = U_{ij} + U_{ijk} \quad (5)$$

in which the 2 body interactions are given by

$$U_{ij} = \frac{H_{ij}}{r_{ij}^{\eta_{ij}}} + \frac{Z_i Z_j}{r_{ij}} \exp\left(\frac{-r_{ij}}{r_{1s}}\right) - \frac{P_{ij}}{r_{ij}^4} \exp\left(\frac{-r_{ij}}{r_{4s}}\right), \quad (6)$$

and the 3 body interactions are governed by

$$U_{ijk} = B_{jik} f(r_{ij}, r_{ik}) \left(\cos\theta_{jik} - \cos\tilde{\theta}_{jik} \right)^2, \quad (7)$$

where r_{ij} represents the distance between i^{th} and j^{th} atom, U_{ij} and U_{ijk} represent the two- and three-body potentials, respectively. Also, r_{1s} and r_{4s} are the interaction cutoff distances. The first, second and third terms on the right-hand side of Equation (6) denote the strength-exponents of steric repulsion, Coulombic interactions and electric polarisability, respectively. In Equation (7), B_{jik} represents the strength of the three-body interaction. Also, θ_{jik} denotes the angle between the vectors r_{ij} and r_{ik} , and $\tilde{\theta}_{jik}$ is constant.

These equations are adequate to computationally analyze the silica aerogel properties in a parallel or pair approach since they are relatively inexpensive and computationally less complex. However, for further rigorous investigations, a multi-body set-up is required to analyze the materials like carbon, which have multiple polymorphing capabilities. To this extent, the strength is parameterized such that it is bound by the local environment conditions using a bond-order function. This is demonstrated by the Tersoff potential and adapted for the silica aerogels by Lei et al. [77]. Ultimately, these approaches can employ molecular dynamics to understand the atomic interactions, pore density, and their impact on silica aerogel's structural and mechanical properties.

The initial research by Kieffer and Angell [65], performed on a non-fractal amorphous silica aerogel, the model is subjected to a rapid isotropic stretching beyond the negative pressure (tensile) until minimum mechanical energy is reached. The structure is allowed to stabilize to determine the geometric and dynamic characteristics. Only a few bonds are ruptured to form voids in the initial expansion stages, which

serve as ‘nucleation sites’ to further bond-ruptures. These voids develop independently at random intervals and locations, dependant on the density of the structure. The final low-density structure resembles an arrangement of a small cluster linked by chains of ions, containing distinguishable voids as shown in Fig. 2.

It shows that the silica aerogels have an affinity for the fractal dimensional structure. Although these systems were modeled with small sizes (300–1500 particles), further studies have been conducted by Nakano et al. [78] in 1993 (41,472 particles). In the recent years, we have performed MD simulations of millions of atoms [47–52]. The fractal dimensions of the lower density aerogels are found to be in the acceptable experimental scattering values. In the later studies performed by Nakano et al. [78], a power-law decay behavior was used to explain the fractality of the aerogels. It was used to determine that the distribution of pore-size occurring in the structure below the tensile limit obeys the power-law decay. In a condensed phase at density 1.6 g/cm^3 , the distribution can be fitted into a Gaussian curve. Similarly, Campbell et al. [79] used the Vasishta potential in 1999 to determine a power-law relation between the elastic moduli densification relative to the density of the aerogels between the range 1.67 and 2.2 g/cm^3 . This power-law exponential is in the range of 3.5 ± 0.2 , which were verified through neutron scattering experiments. Moreover, the power-law relationship between the bulk and shear moduli with the density was proposed.

In 2010, Murillo et al. [80] successfully continued the research to determine the mechanical characteristics through molecular dynamics simulations by obtaining results that agreed with the experimental values of Gross and Fricke [81] and Woignier et al. [82,83]. In this study, the preparation of the sample was carried out by increasing and decreasing the temperature of the system in a stepwise manner. Since the atoms in the system are initially farther from the equilibrium state, it enabled them to form any kind of structure. This study was conducted with a vast density range of 0.23 – 2.2 g/cm^3 , and provided an overview of the silica aerogel’s mechanical properties as part of comparative research against the experimental values obtained by Woignier et al. [83] and Campbell et al. [79]. The fractal dimensions were calculated to fall in the range of 2.33 ± 0.23 , which were verified by the practical

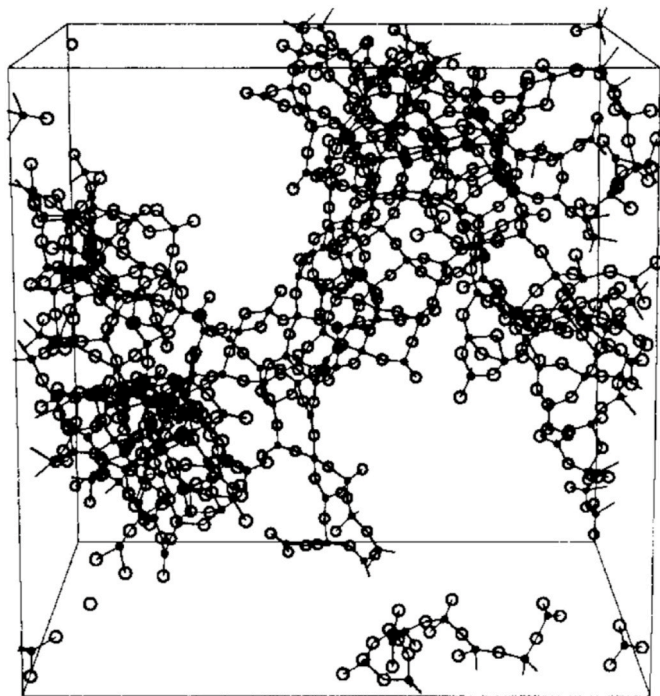


Fig. 2. Three-dimensional (3D) Molecular arrangement of silica aerogel as demonstrated by Kieffer and Angell [65] (Reprinted with permission from Ref. [65]. Copyright 1988 Elsevier.).

values obtained by Courtens and Vacher [84]. The elastic modulus of the glass aerogel was found to be around 70 GPa , which falls approximately in the same range provided by the experimental results obtained by Muralidharan et al. [85]. The obtained results showed the power-law dependency of elastic modulus and tensile strength with bulk density (see Fig. 3). The exponent values were the range of 3.11 ± 0.21 and 2.53 ± 0.15 for elastic modulus and tensile strength, respectively [79,81–83].

In 2013, Liu et al. [86] validated the results from Murillo et al. [80] by adopting a two-tier simulation model - one with the main backbone structure and the other with a nanoporous structure. A prominent result of this study is the determination of thermal conductivity, which was dependent on the density. Moreover, in 2013, Lei et al. [77], provided an improved version of the negative pressure rupturing technique to create the molecular models and extended it to study the elasticity modulus and Poisson’s ratio’s dependency on the density. For this purpose, the model was subjected to a uni-axial and bi-axial tensile loading. The outcome obeyed the power law for density with the exponents 3.378 and -0.107 for Young’s modulus and Poisson’s ratio, respectively. If the silica aerogels are assumed to be homogeneous and isotropic, then the constitutive equations of elastic materials hold good, with only Young’s modulus and Poisson’s ratio being the variable parameters.

Ferreiro-Rangel and Gelb [87] investigated the bulk structure of the silica aerogel to gain an insight into the bulk mechanical characteristics, wherein two different approaches were utilized, fluctuation analysis and compression/expansion simulation techniques. It was observed that the bulk modulus dependence on density follows a power law with an exponent in the range of 3 and 3.15 , which was well supported by the experimental work of Scherer et al. [88] (see Fig. 4). Also, Hou et al. [89] studied the adsorption capability of reinforced silica aerogels for their adsorption capability and their dependence on the relative pressure and the reinforcement material. In addition, the study also noted the impact of the reinforcement material on the mechanical properties, such as the elastic modulus and compressive strength.

Another noteworthy approach to obtain the MD simulation model includes a series of heating, quenching and instantaneous expansion process, which was discussed in detail in our previous work Patil et al. [70] (see Fig. 1). The structural characterizations and mechanical properties of silica aerogel achieved from this approach are in excellent agreement with experimental studies in the literature. In this work, the MD simulation system was built of $192,000$ atoms, and it was subjected to tensile, compression and cyclic loading tests. The resulting Young’s modulus was shown to express a power-law relation against the density. The exponent value was estimated to be 3.25 ± 0.1 , which is well in the range of previous experimental and computational studies [69, 90–94]. Fig. 5a shows that the low-density aerogels were more ductile, while the higher density ones were brittle under tensile loading. Moreover, under compression, low-density aerogels deformed without a significant increase in stress (see Fig. 5b). On the other hand, higher density aerogels offered a much more prominent resistance to deformation resulting in stiffer behavior and early densification.

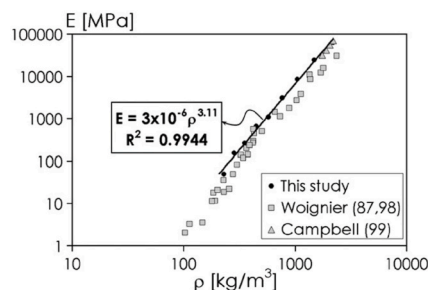


Fig. 3. Comparison of the simulation results of Young’s modulus in the work of Murillo et al. [80] to the results obtained by Woignier et al. [82,83] and Campbell et al. [79] (Reprinted with permission from Ref. [80]. Copyright 2010 Elsevier.).

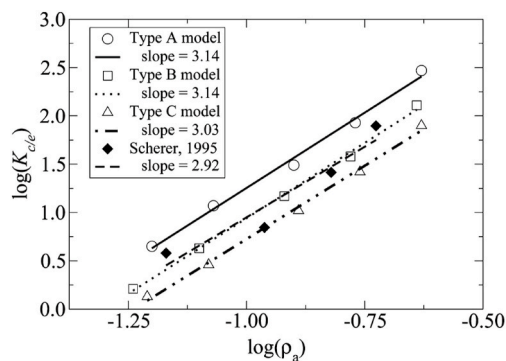


Fig. 4. Bulk modulus of silica aerogel as obtained by Ferreiro-Rangel and Gelb [87] (Adapted with permission from Ref. [87]. Copyright 2013 American Chemical Society.).

Furthermore, cyclic tests were conducted, which offered new insights into the inelastic effects such as cyclic stress softening and residual strains (see Fig. 5c). Novel relations were proposed that describe the growth of residual strains, dissipated energy and the average tangent modulus as functions of applied maximum strain per cycle. These new relations give a comprehensive insight into the damage characterization and criteria for the silica aerogel.

Goncalves et al. [95] performed MD simulations on the large sample sizes. In this work, atomic interactions of silica were modeled using the Wolf BKS potential (van Beest, Kramer and van Santen), which was proposed by Carré et al. [96]. The main modification compared to the BKS force field [97] was the addition of a cut-off on long-range electrostatic interactions. The study concluded that the mechanical results obtained, such as the elasticity modulus and fractal dimensions, using the large size model, concur with the previous experimental and computational results [70,82,83,92,94]. Goncalves et al. [95] also studied the hysteresis, which was obtained by subjecting the model to an uni-axial loading. Investigated under four different strain rates with several uniaxial tensile load-unload cycles were performed. The hysteresis loops are shown in Fig. 6. It can be observed that the higher the strain rates, the smaller is the hysteresis loop.

According to Hüsing and Schubert [98], the nanoporous structure of silica aerogel implies a low load-bearing capacity. Despite many currents and potential applications of silica aerogels, one major concern is their failure in a brittle manner, which depicts a sudden, strong, swift cracking of material with little or no evidence of plastic degradation prior to the occurrence of the fracture. A thorough investigation of the crack propagation and its effect on the resulting fracture strength, fracture toughness and strain energy release rate remains to be examined. Therefore, recently, MD simulation was used to study the fracture behavior of silica aerogel (see Fig. 7), which has been used as a powerful tool to elucidate the crack initiation and propagation in brittle materials at the nanoscale [47].

We found that the fracture properties, such as the strain energy release and fracture toughness, increase with the crack depth [47]. Moreover, the fracture toughness, as well as strain energy release, show a power-law relationship with density, and the exponent of these relations were 2.02 ± 0.05 and 1.16 ± 0.04 , respectively. This study is beneficial for the selection of silica aerogel for the specific application [47].

The measurement of mechanical properties of highly nanoporous materials is difficult, which is mainly due to a low load-bearing capacity [98]. Hence, Patil et al. [49] used the nanoindentation technique to measure the elastic and inelastic behaviors on the nanoscale using all-atom simulations on large samples. Moreover, the obtained MD simulation results are compared with experiments. Due to the highly randomized solid skeleton structure of silica aerogel, the new approach has been proposed to compute the projected contact area.

In Fig. 8a, the elastic modulus obtained from MD simulations shows stiffer nature compared to experiments as well as literature values [69, 70,87,95,99–104]. Fig. 8b shows the comparison of hardness values obtained from MD simulations, experiments and Chen and Ke [105]. The hardness values are calculated using the true projected area. Therefore, the MD simulations show higher values because those represent the actual local contact pressure, especially, low densities silica aerogels.

Recently, MD simulations were carried out to determine the shock

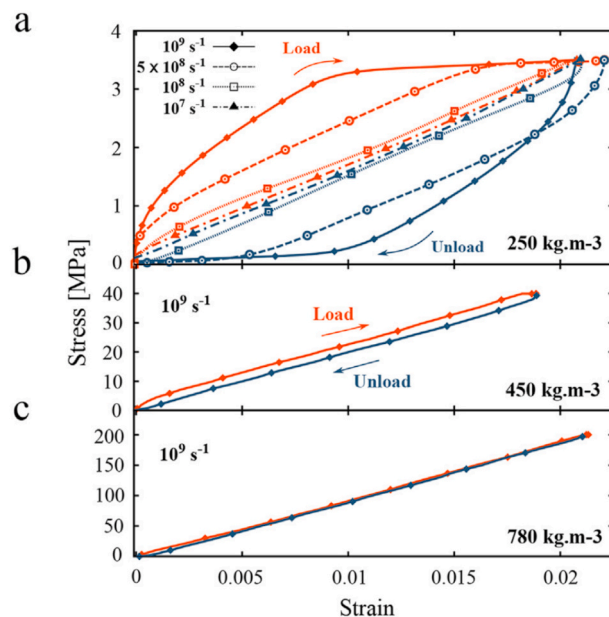


Fig. 6. To study the hysteresis behavior of silica aerogel of 803 nm³ sample at 300 K temperature, uniaxial tensile load-unload cycles carried out on different densities: (a) 250 kg m⁻³, (b) 450 kg m⁻³ and (c) 780 kg m⁻³ (Reprinted with permission from Ref. [95]. Copyright 2018 Elsevier.).

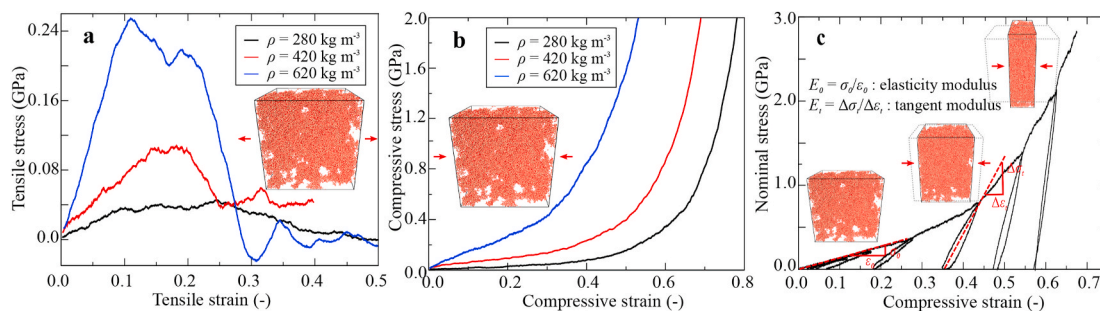


Fig. 5. (a) Stress-strain curves for silica aerogels from MD simulations under tension, (b) compression and (c) consecutive compressive loading and unloading cycles (Adapted with permission from Ref. [70]. Copyright 2017 American Chemical Society.).

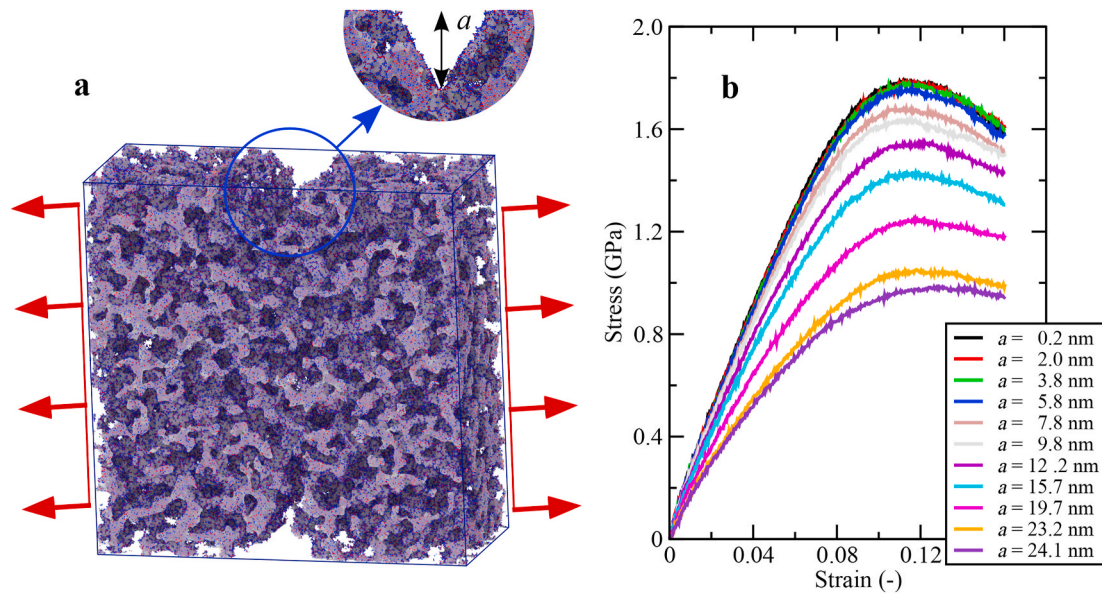


Fig. 7. (a) Schematic representation of an all-atom model and rupture of silica aerogel of a density of 645 kg m^{-3} with a (center) denotes the notch depth. (b) Stress-strain curves obtained from MD simulations of the silica aerogel models of 1155 kg m^{-3} density for varying notch depths (Reprinted with permission from Ref. [47]. Copyright 2018 Elsevier.).

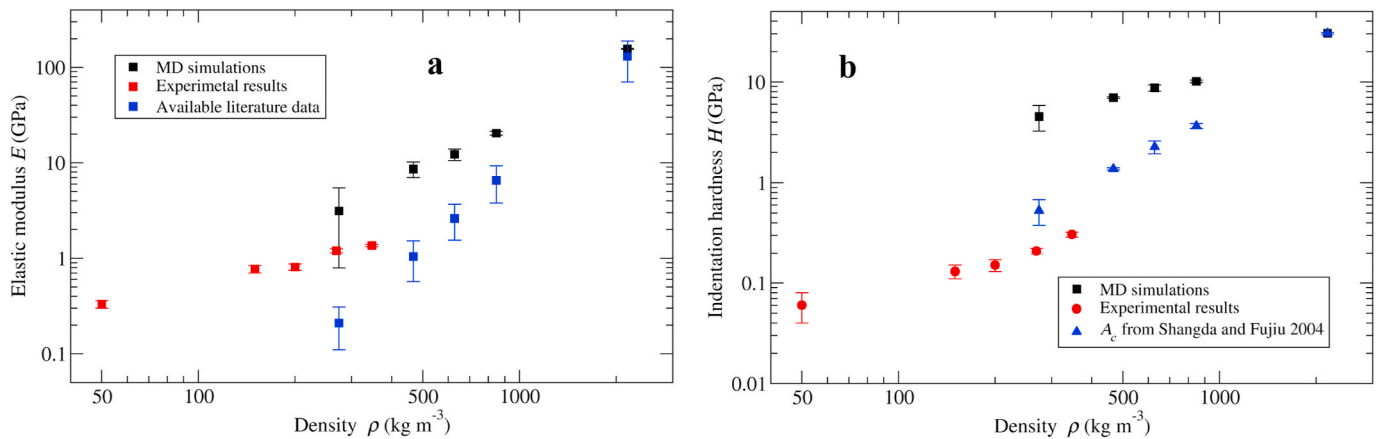


Fig. 8. The results obtained of (a) Young's modulus and (b) hardness as a function of density in comparison with available literature values (Reprinted with permission from Ref. [49]. Copyright 2019 Elsevier.).

response of silica aerogel by Patil et al. [48]. The objective was to study the shock velocity generated due to the particle velocity and its effects (see Fig. 9a). In this work, the silica aerogels of density $209\text{--}847 \text{ kg m}^{-3}$ were considered. The shockwave studies were performed by implementing the reverse ballistic approach. The particle velocity was noted to increase linearly with the generated shock velocity, as shown in Fig. 9b. The linear slopes were found to be in excellent agreement with the experimental and theoretical works [106–110]. Moreover, it was found that for the low-density silica aerogels, the von-Mises stress values increased at a slow rate and the maximum value reached was found to be lower than that of higher density aerogels. The von-Mises value behaves non-linearly and increases with the progress in shock front. This is because of the densification behind the shock front.

Another prominent work in MD simulations was done by Morthomas et al. [111]. In this study, the thermal conductivity was investigated using a stable non-equilibrium molecular dynamics (NEMD) simulation technique. Fig. 10 shows the thermal conductivity of silica aerogel as a function of the density. It utilizes the diffusive nature of heat transfer to effectively observe and compare the thermal conductivity from the MD simulations to that calculated from differential equations. The thermal

conductivity during the transient states is predicted for different system sizes and densities.

Due to poor mechanical properties, low-density silica aerogels become limited to some of the specific applications. Therefore, many researchers are working on how to enhance the mechanical properties of silica aerogel. The most prominent and accepted approach is reinforcing the silica aerogel by adding nanoparticles, polymers, or fibers. Recently, carbon-based nanomaterials, such as carbon nanotubes, carbon fibers and graphene, are being used as reinforcement materials in silica aerogels. For the first time, silica aerogel nanocomposites were investigated rigorously using MD simulations by Patil et al. [51,52]. The work provided new insights into the fundamental understanding of how the reinforcing materials, such as glass fibers, graphene sheets, and carbon nanotubes, into a silica aerogel matrix, improve the mechanical properties and the deformation behavior due to the already existing voids at the nanoscale [51,52].

There have been many varied and excellent works that have contributed to the development of the applications of molecular dynamics simulations towards the understanding of silica aerogels' behavior and properties at the nanoscale. Another method to analyze the

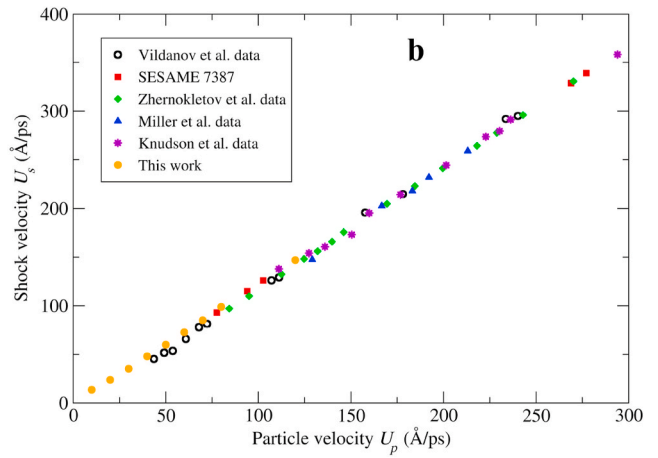
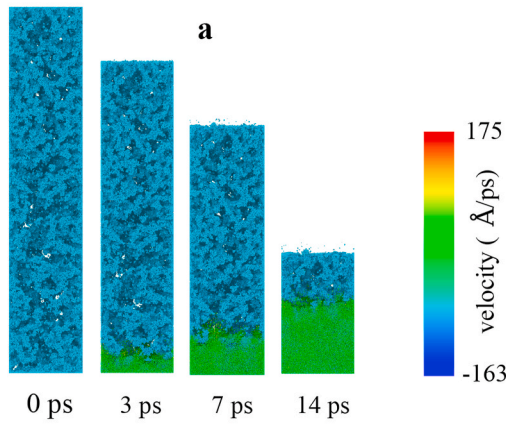


Fig. 9. (a) Study of shock velocity propagation in the silica aerogels of density of 467 kg m^{-3} at particle velocity of 70 Å ps^{-1} [48]. (b) Shock Hugoniot of silica aerogel of density of 209 kg m^{-3} in comparison of experiments performed in the literature [48,106–110] (Reprinted with permission from Ref. [48]. Copyright 2019 Elsevier.).

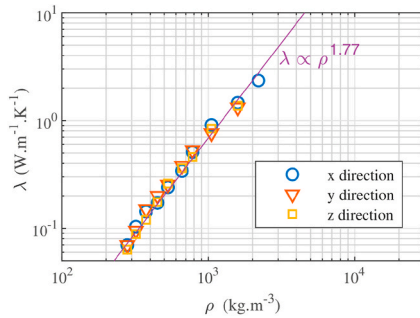


Fig. 10. Thermal conductivity as a function of density of silica aerogel (Reprinted with permission from Ref. [111]. Copyright 2019 Elsevier.).

mechanical properties of silica aerogels is coarse-grained modeling, which shall be discussed in the next section.

2.2. Coarse-grained modeling

The Coarse-grained models are used to gain a better understanding of the linear elastic properties of certain materials. Usually, they are applied to molecular structures where groupings or clusters are present. These groupings are represented as a single entity in the coarse-grained models, thereby reducing the complexity and simulation time of the calculation. These models were first introduced in the 1970s by Levitt et al. [112], where they were introduced to reduce the complexity of the protein structures. This implies that there will be a loss of atomic clarification that is presented by the molecular dynamics. Coarse-grained models for the silica aerogels can be broadly divided into two categories based on the calculation involved and the algorithm used. They are hard-sphere aggregates and flexible coarse-grained models. The hard-sphere aggregates models follow a methodology in which, the non bonded particles interact like hard spheres or “billiard balls”, while bonded particles are rigidly held together [53]. The best examples for hard-sphere aggregates models are Diffusion Limited Aggregation (DLA), Reaction Limited Aggregation (RLA), Diffusion Limited Cluster Aggregation (DLCA) and Reaction limited Cluster Aggregation (RLCA). Both DLA and RLA are models in which clusters grow by the addition of monomers. However, the difference is, in DLA, the reactions have no barriers. Hence the clusters grow by diffusion. On the other hand, RLA is modeled based on the fact that not all collisions by the addition of monomers need to react. DCLA and RCLA are models in which clusters are themselves mobile, where they meet and aggregate each other in

order to create reactions.

Hasmy et al. [113] were the first to adapt DLCA to the silica aerogels in 1993, in which the DLCA silica aerogel model results were compared with the short and long-range results of small neutron scattering (SANS) spectra of colloidal silica aerogels. Diffusion limited, ballistic, limited reaction aggregates, with and without restructuring, were used to enable single aggregates. Restructuring provided results that were highly compact but similar fractals. Though only qualitative, DLA was found to provide the best outcomes that were most agreeable with the experimental results. In 1994, Hasmy et al. [114] used periodic boundary conditions (PBC) to model the DLCA model. In this model, the clusters aggregated themselves and were chosen by considering the inverse of the effective diffusion constant. This measure was undertaken to ensure that the larger clusters diffuse more slowly than the smaller ones. The DLCA model results fit very well in nearly all aspects of the colloidal aerogels but not quite well with the basic and neutral aerogels.

This work was further extended by Ma et al. [115,116] in the early 2000s to explain the compliance of the silica aerogels. In this study, the DLCA was combined with the ‘dangling bond deflection’ (DLCADef) to provide an extensive loop structure. According to Ashby and Gibson [117], theoretically, the exponential should converge to a value of 2. However, experimentally, it was observed to be in the range of 3 and 4. The reason for this was considered to be the unique structure of the silica aerogels, which consisted of dead-ends. These dead-ends are clusters, which are connected to the backbone of the aerogels network at only a single point. The study contradicted this hypothesis by demonstrating a DLCADef model, which did not have many dead-ends. It was observed that rather than dead-ends, loops provide the necessary stiffness in the aerogels. The study also provided an almost perfect fit bulk modulus exponential of 3.6, as shown in Fig. 11.

In 2002, Ma et al. [118] further extended this model by providing detailed mechanical characteristics. In this work, the structure is treated as a beam, and the deformation factors were analyzed. It can be observed that the maximum deformation occurs due to the shear bending of the structure. The study shows that the total strain on the network does not contribute to the exponential conundrum in any way. This implies that this model is not appropriate to study the yield and compression hardening of gels. The study shows the dependence of Poisson’s ratio with the silica aerogel density by considering bond compliance factors on the DLCADef network (see Fig. 12). The study explains that due to the nature of the aggregation process, the density variation in the gels is accompanied by a prominent change in the connectivity, unlike in the polymer or ceramic foams. This is why the exponential compliance of the silica aerogel does not correspond to the value predicted by the open-cell foam model.

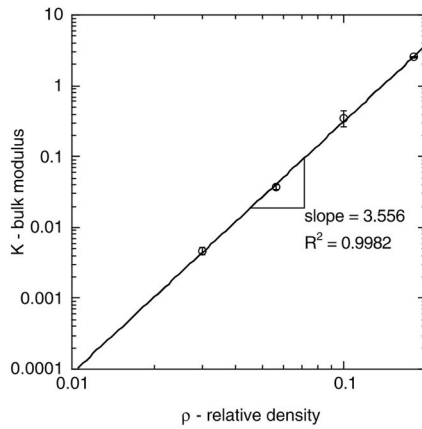


Fig. 11. Bulk modulus of silica aerogel determined by Ma et al. [116] (Reprinted with permission from Ref. [116]. Copyright 2001 Elsevier.).

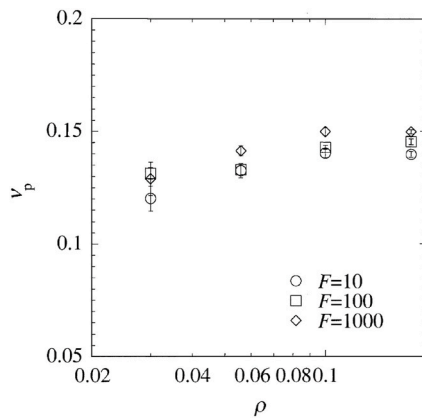


Fig. 12. Variation of Poisson's ratio of silica aerogels with respect to relative density ρ of the DLCADEF networks at different bond compliance factor F (Reprinted with permission from Ref. [118]. Copyright 2002 Elsevier.).

In 2007, Gelb et al. [104] used a fully flexible model that accounted for the bond breakage and creation upon the gel structure's deformation. The salient feature of this study was the incorporation of the bond creation factor P . The study shows that at sufficiently low P , the simulations achieve a scaling behavior, in which the final gel structure is mostly independent of P . Hence, it can be used to compare with the experimental results. Along with the scaling factor, the study also evaluates the impact of varying viscosity of the sol-gel on the model, as shown in Fig. 13.

Since these models are used mainly because of their high efficiency in providing the linear elastic properties of the materials, they are mostly applicable in the initial stages of the evaluation. For further, in-depth calculations, these models can be combined with other models, such as finite element method (FEM).

2.3. Continuum mechanics modeling

The continuum models incorporate models, such as meso and micromechanical multiscale models, which in turn encompass a heavy involvement of FEM. These models are mainly used to analyze the stress-strain effect of the load on the structure. These methods may be used in combination with molecular dynamics or coarse-grain models to provide a comprehensive overview of the behavior. In 1993, Gross et al. [119], used ultrasonic testing to analyze the ultrasonic behavior of the silica aerogels. In this study, the velocity and attenuation of ultrasonic waves were measured as a function of frequency, density, the gas

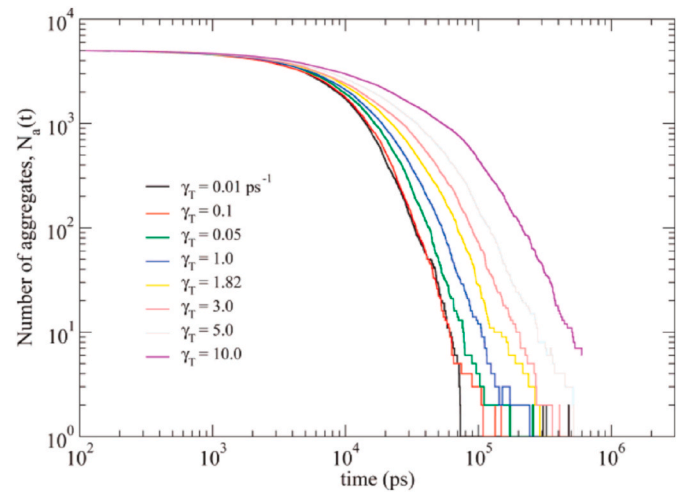


Fig. 13. Number of aggregates versus time at different viscosities (Adapted with permission from Ref. [104]. Copyright 2007 American Chemical Society.).

pressure in the pores and elastic strain. The density is considered as one of the most critical parameters that influence the sound velocity and hence the elastic modulus. The study also observed that the aerogels indicate an unusual non-linearity, which is highly dependent on the density, irrespective of their composition. Although the study does not shed light on the attenuation and creep mechanisms, it has opened new avenues for research in the mechanical properties.

Lu et al. [120] proposed that a two-tier model approach, which can be better suited to analyze the mechanical properties of the silica aerogel composites. This is done by creating an elastic contact model with a finite bond size and getting the simulation parameters through MD and modified DLCA computations. The model itself is generated using a modified EET technique by using interphase between the FEM one-dimensional (1D) elements acting as a fiber matrix and the applied load. Though a complex computation procedure, this technique proves to be a promising approach to quantify the mechanical properties of silica aerogels. Fig. 14 shows the influence of the fiber volume fraction on Young's modulus and tensile strength of the silica aerogels. It can be noted that Young's modulus increases approximately linearly with the fiber volume fraction.

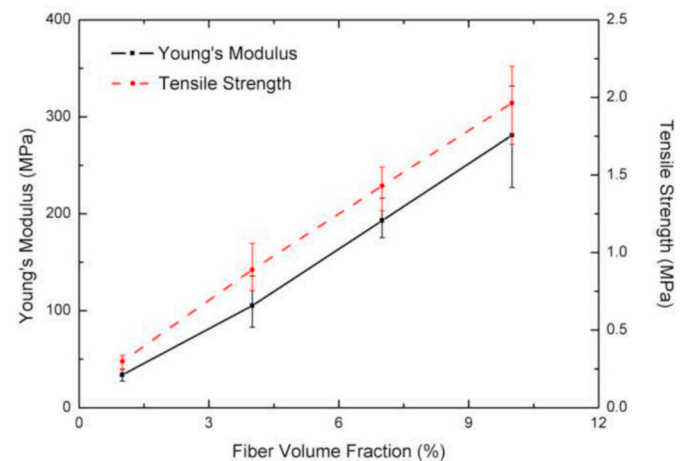


Fig. 14. Variation of Young's modulus and tensile strength due to change in the fiber volume (Reprinted with permission from Ref. [120]. Copyright 2015 Elsevier.).

2.4. Modeling of silica aerogels nanocomposites

Applicability of silica aerogels has remained very limited due to the poor mechanical characteristics, such as brittleness and low tensile strength. Therefore, the reinforcement of the silica aerogels has been considered a prominent approach to enhance their mechanical characteristics. Experimentally, composites of silica aerogel were synthesized using carbon nanotubes [121,122], carbon nanofibers [123,124], carbon aerogels and graphene [125,126]. These composites were extremely lightweight and highly porous, still possess high tensile strength and toughness. Moreover, glass fiber was used as a reinforcement candidate in silica aerogels to enhance the tensile strength, flexibility and thermal insulation properties [127–131].

Recently, the first MD simulation study was performed by Patil et al. [51]. In this work, glass fibers, carbon nanotubes and graphene sheets were used as reinforcement candidates into silica aerogels using MD simulations. Moreover, the mechanical behavior of these nanocomposites due to the pre-existing cracks was studied. The outcome of this work shows that the overall mechanical properties were significantly improved with the addition of reinforcement material. In particular, the mechanical properties of carbon-based nanocomposites were remarkably enhanced than pure and glass fibers reinforced silica aerogels (see Fig. 15). Moreover, along with the longitudinal axis loading, Young's modulus of the glass fibers, carbon nanotubes and graphene sheets reinforced silica aerogel nanocomposites were ~ 3.5 , ~ 11.5 and ~ 9.5 times higher than pure silica aerogels, respectively. Furthermore, for the influence of crack depth, it is concluded that the reinforced candidates are mainly responsible for carrying the tensile load.

In our previous work, the reinforced materials were aligned in one direction [51]. Therefore, the extreme limits of the enhancement in the mechanical properties were investigated. The further extension of this work was to consider the random distribution of reinforced materials in the silica matrix [52]. In this work, glass fibers of different length to diameter (l/d) ratios and different weight percentages were randomly distributed in silica aerogels to investigate their influence on the mechanical properties (tensile strength, elastic modulus and compressive behavior) using MD simulations. The study concluded that due to axial and bending deformation of the reinforced glass fibers, the proposed nanocomposites possessed significantly improved mechanical properties compared to pure silica aerogel (see Fig. 16). Recently, we also studied the silica nanocomposites using a multiscale bottom-up approach, wherein MD simulations were used at the nanoscale to obtain the mechanical parameters, and continuum modeling was used to perform the 3D macroscopic network behavior [132]. The outcome of this work

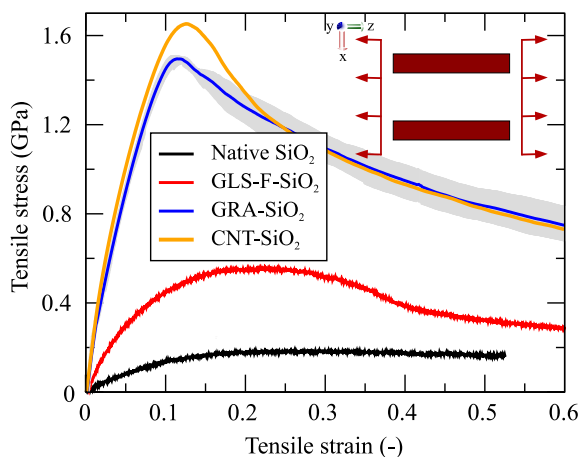


Fig. 15. Comparison of tensile stress-strain curves of pure silica aerogel and its nanocomposites. Here, tensile loading direction was along the longitudinal axis (stress in z-axis). The standard error deviation is indicated by the gray shaded area (Reprinted with permission from Ref. [51]. Copyright 2020 Elsevier.).

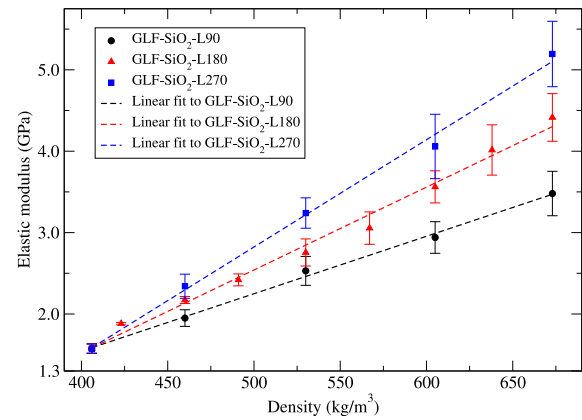


Fig. 16. Comparison of the elastic moduli of native silica aerogels and their nanocomposites with standard error bars (Reprinted with permission from Ref. [52]. Copyright 2020 Elsevier.).

shows that the obtained parameters from MD simulations to the micro-mechanical model provide a physical estimation. This approach can be effectively used for other types of nanocomposites.

In summary, silica aerogel has high porosity, low thermal conductivity, high surface area, and good chemical stability. However, it is brittle in nature. The elasticity moduli display a power-law dependence on the density, with the exponent estimated to be 3.25 ± 0.1 [70]. In the compression tests, silica aerogel behaved like polymeric foams. The compression stress-strain curves were divided into three regions: a linear elastic region (< 0.04 strain), a yielding region, and a densification region. For flexible polyurethane foam, relative Young's moduli in compression vary linearly with relative density [133]. The silica aerogel's thermal conductivity is 0.032 W/m K , which is lower than polyurethane (0.25 W/m K) at room temperature [134]. Moreover, Chang et al. studied the thermal conductivity of silica aerogels/polyurethane composites for their various compositions [134]. In MD modeling of low-densities silica aerogel ($< 220 \text{ kg m}^{-3}$), the skeleton connections throughout the models were not stable. Therefore, in the literature, many MD simulation studies were performed on the densities of $> 220 \text{ kg m}^{-3}$. In the future, one can carry out MD simulations of the low-densities silica aerogel using all-atom simulations. The resultant properties of these simulations can be directly compared with experiments. The thermal properties of the silica aerogel nanocomposites (reinforced with glass fibers, carbon nanotubes, and graphene) can be studied using MD simulations. Moreover, the Lattice Boltzmann Methods (LBM) can be used to investigate the air-flow in silica aerogels.

Like the silica aerogels and their nanocomposites, the other aerogels have garnered significant interest in the research community. In the following sections, the different mechanical modeling approaches for other types of aerogels are listed and also critically discussed.

3. Carbon and carbon nanotube aerogels

Carbon aerogels were first invented by Pekala et al. [135] in 1990 via pyrolysis of resorcinol-formaldehyde-based organic aerogel. As one of the recently invented material, the literature is not as extensive as silica aerogels. In the past few decades, with increasing interest in carbon, as it is one of the lightest materials, be it in the form of fiber, crystalline or amorphous forms, various researchers have taken an interest in experimenting and developing the applications for carbon aerogels. In fact, in 2013, the carbon-based aerogels clinched the title of "lightest material" from the silica aerogels with a density lesser than $\mu\text{g/cm}^3$ [136]. They are manufactured to have high surface areas with good electrical conductivity and mechanical properties [137]. There have been several experimental protocols customized exclusively to study the carbon aerogels [138–142]. Of particular interest are the carbon nanotube

(CNT) based aerogels which have superior and enhanced mechanical properties. This is because an opportunity arises to improvise the mechanical properties when the CNTs are cross-linked with the carbon sol-gels. Several experimental results have already accredited to the enhanced properties of these carbon nanotube-based aerogels [135, 143–147]. Since these aerogels were discovered relatively recently, the computational results and research papers are also relatively few. This review makes an effort into categorizing the available research papers on computational analysis.

3.1. Molecular modeling

Recently, the AIREBO force-field has been widely used to model the C–C (carbon–carbon) interactions in CNTs and graphene aerogels [148, 149]. The interatomic potential energy is given by

$$U_{\text{AIREBO}} = \frac{1}{2} \sum_i \sum_{j \neq i} \left[U_{ij}^{\text{REBO}} + \sum_{k \neq i} \sum_{l \neq i,j,k} U_{kijl}^{\text{TORSION}} + U_{ij}^{\text{LJ}} \right]. \quad (8)$$

Here, U_{AIREBO} is the overall potential energy of MD system, U_{ij}^{REBO} is the REBO term, which describes potential between covalently bonded atoms, $U_{kijl}^{\text{TORSION}}$ is the torsional part of the potential, which depends on the dihedral angle, and U_{ij}^{LJ} is the Lennard-Jones potential.

In 2001, Gavalda et al. [150] made a concentrated effort into performing a molecular simulation study, which evaluated the nitrogen adsorption capability of the carbon aerogels. Although they relegated their studies to the mesoporous and microporous structures, they laid a fundamental groundwork for future molecular study. This research made use of the Reverse Monte Carlo method to ascertain the values. The molecular adsorption representation and isotherm values can be seen in Fig. 17. One can observe, the simulation values corresponded well with the experimental values obtained through X-ray diffraction. This method is not as preferred while simulating carbon nanotube aerogels as silica aerogels because of the fibrous nature of carbon. Hence, most of the mechanical properties available as yet are through finite element modeling and experiments, as discussed in the next subsection.

3.2. Continuum modeling

Yang et al. [151] provided a more in-depth insight into the properties of carbon aerogels using the continuum models. In this work, experiments and simulations were performed to demonstrate the electro-sorption of ions in an aqueous solution by carbon aerogel electrodes by creating an electrical double layer model. This study gives an understanding of the carbon aerogel's electrosorption capacity, which can be used to determine the use of these materials as electrical insulators. The formation of these models is quite elaborate. Still, once created, it can be used for variable parametrization to verify capacitance based on the concentration of ions in the aqueous solution and voltage, as shown in Fig. 18.

Hemberger et al. [152] performed both experimental and numerical analyses on functionally graded carbon aerogels to ascertain the thermal conductivity of the CNTs. During the study, the thermal conductivity as a function of pore diameter and time were evaluated. It can be observed that these aerogels are inherently good thermal insulators, although the

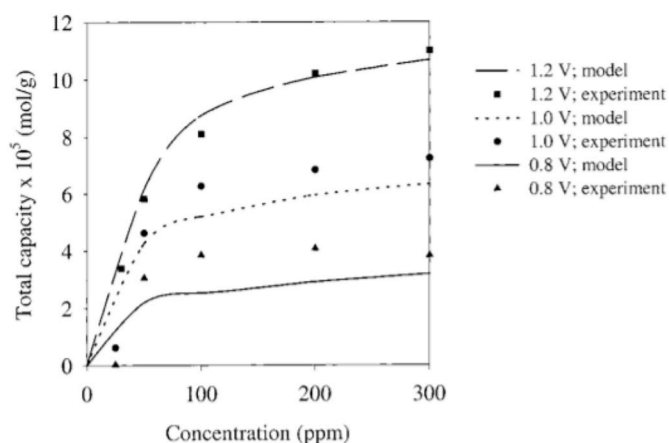


Fig. 18. Electrosorption capacity as a function of ion concentration in aqueous solution (Adapted with permission from Ref. [151]. Copyright 2001 American Chemical Society.).

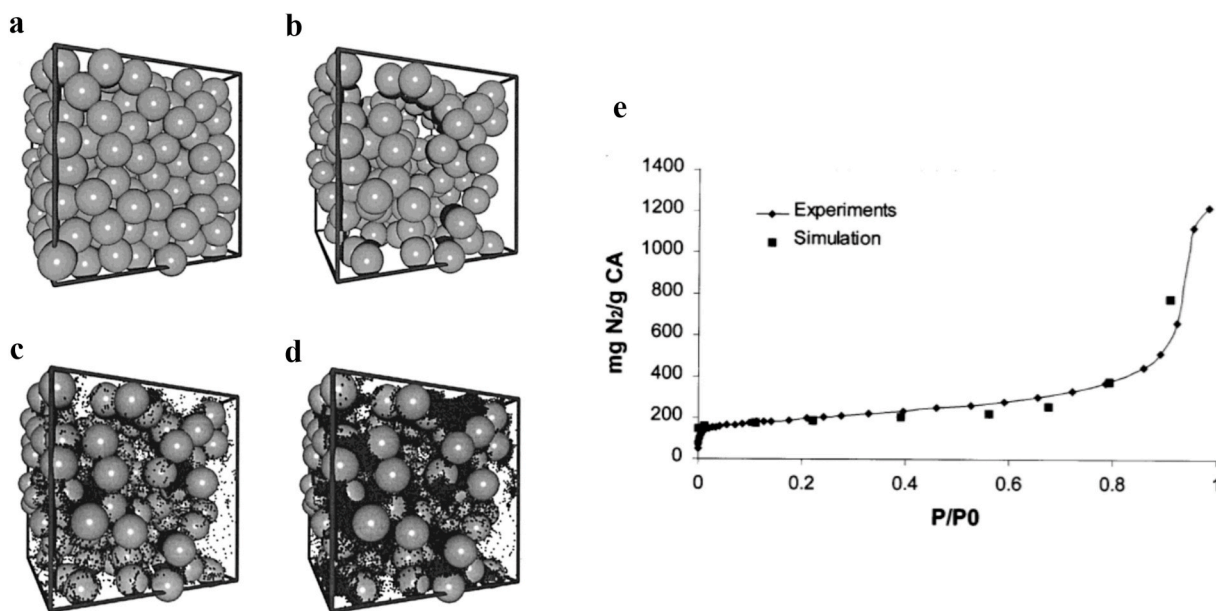


Fig. 17. Molecular Arrangement of carbon aerogel. (a) Close-packed configuration and (b) final configuration. Nitrogen adsorption in carbon aerogel at relative pressures of (c) 0.56 and (d) 0.79. (e) Comparison of simulations and experiments of nitrogen adsorption isotherms at 77 K (Reprinted with permission from Ref. [150]. Copyright 2002 Elsevier.).

results did not coincide with that obtained from the classical approaches [153,154]. A plausible reason was due to the anisotropy of the material. Recently, Gong et al. [155] created a 3D simulation model to predict the heat conduction thermal resistance of CNT aerogels in various gaseous media by using an off-lattice Monte Carlo method. Single-walled CNTs (SWCNTs) are simulated as a 3D mesoporous model, as shown in Fig. 19a. The red and green surfaces symbolize the heated and cooled surfaces, respectively. The estimation of the thermal boundary resistance (TBR) between the aerogels and four permeated gases are compared with the experimental results obtained by Schiffrès et al. [156] (see Fig. 19b). It is observed that the TBR decreases with an increase in molecular mass. The study concludes that aerogels with higher SWNT volume fraction, longer length, and smaller diameter results in a higher effective thermal conductivity (see Fig. 19c).

With the help of these simulation results and various experiments, the carbon aerogels have been used in the industry as good energy storage systems [157–159], good thermal and electrical insulators and lightweight devices like satellites.

Carbon aerogel is one of the lightest materials with high surface areas with good electrical conductivity and mechanical properties. The lowest thermal conductivities of carbon aerogel were varied from 0.0263 to 0.054 W/m K [160,161]. These values are lower than polyurethane (0.25 W/m K) at room temperature [134]. As silica aerogels have been used for high-energy absorption applications, carbon and carbon nanotube aerogels can be investigated for high impact applications using MD simulations. Moreover, nanocomposites of carbon and carbon nanotube aerogels can be studied numerically, which can be helpful in producing a tailor-made design for specific applications. Therefore, further exploring the computational modeling and simulations could lead to various other applications.

4. Graphene aerogels

Graphene, another form of carbon, has been used to create the lightest solid material in the world – Graphene aerogel or aerographene. In 2010, Worsley et al. [162] reported the first graphene aerogel with covalent cross-links with a framework that incorporated the carbon junctions. Later on in 2013, the research team lead by Gao Chao [136], created a 3D aerographene with the lowest density by freeze-drying a solution of carbon nanotubes and graphene, resulting in a sponge-like solid. As one of the newest solid materials discovered, the literature on mathematical modeling of graphene aerogels is even lesser than the carbon aerogels. However, a few experimental works have been carried out [163–173].

4.1. All-atom modeling and simulations

The molecular dynamics models of aerographene are extremely rare.

This is because of the two-dimensional (2D) nature of graphene. Recently, an effort was made by Ma et al. [174] to create a molecular dynamics model of both graphene aerogel (GA) and graphene hydrogel (GH) simultaneously as a comparative measure. This study aimed to prove the enhanced capability of these materials to act as water pollutant filters by demonstrating their adsorption capabilities. From this study, it is evident that the graphene hydrogels would be most effective in the adsorption of oil from water being less harmful to the environment. One can also ponder whether the aerographenes would be similarly best suited for the adsorption of air pollutants. Qin et al. [175] constructed the 3D graphene aerogel model in MD simulations by mimicking the synthesis procedure. This work mainly focused on the mechanical properties, such as tensile strength, compression strength, buckling modulus and Young's modulus. This study conclusively displays that the graphene aerogels are ten times stronger than mild steel. However, a density of graphene aerogel is 4.6% that of mild steel.

Recently, we have investigated the mechanical and fracture properties of graphene aerogel [176]. This was the first of its kind of study, wherein the deformation behavior and properties of graphene aerogels under the cyclic compression loading and tensile loading of pre-existing cracks were investigated. Fig. 20 shows the schematic representation of the construction of the MD simulation model. Here, the AIREBO potential for C–C and Lennard–Jones (LJ) potential for carbon-water and water-water were used in MD simulations [148,149]. Fig. 21a and b demonstrate the comparison of the elastic modulus (E) versus density (ρ) and tensile strength (σ_{ts}) versus density (ρ) achieved in this work with the previous works in log-log scale. This work concluded that power-laws could be used to describe the relationship between the above properties. For E versus ρ , the computed exponential value was 2.95 ± 0.05 , which is in excellent agreement with the literature studies [169,175,177–181]. For σ_{ts} versus ρ , the exponent was 1.61 ± 0.04 , which is in good agreement with the literature studies [169,175,178].

Fig. 22 shows the effects of a/h on σ_f and K_{IC} . As the a/h ratio increases, the σ_f values decreased (see Fig. 22a). However, for a considered range of the a/h ratio, the K_{IC} values were approximately constant for all densities (see Fig. 22b). Moreover, a power law can be expressed to describe the relationship between K_{IC} and ρ with the exponent of 1.41 ± 0.04 .

4.2. Continuum modeling

Even though graphene aerogels are one of the most recent materials to be discovered due to the high interest, applicability and experimentation, reinforced graphene aerogels have been created depending on the application. Hence, most of the simulations consist of a reinforced GA to better understand the effects. For instance, in the study conducted by Fan et al. [182] in 2015, GAs have been reinforced with Polymethyl methacrylate (PMMA) to form a composite and then evaluated for

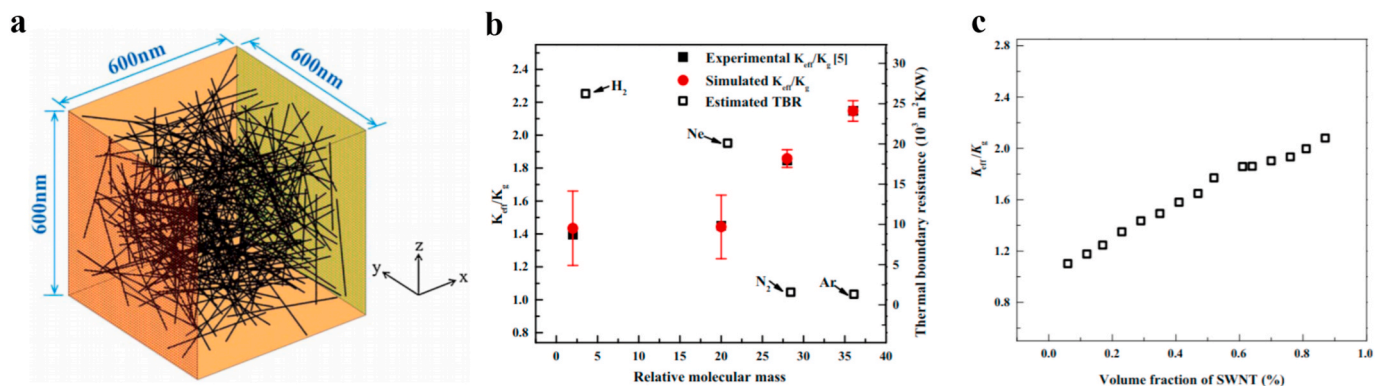


Fig. 19. (a) The 3D molecular arrangement of SWCNTs. (b) Effective thermal resistance as a function of molecular mass. (c) Dependence of effective thermal resistance on the volume fraction of SWCNT (Reprinted with permission from Ref. [155]. Copyright 2015 Elsevier.).

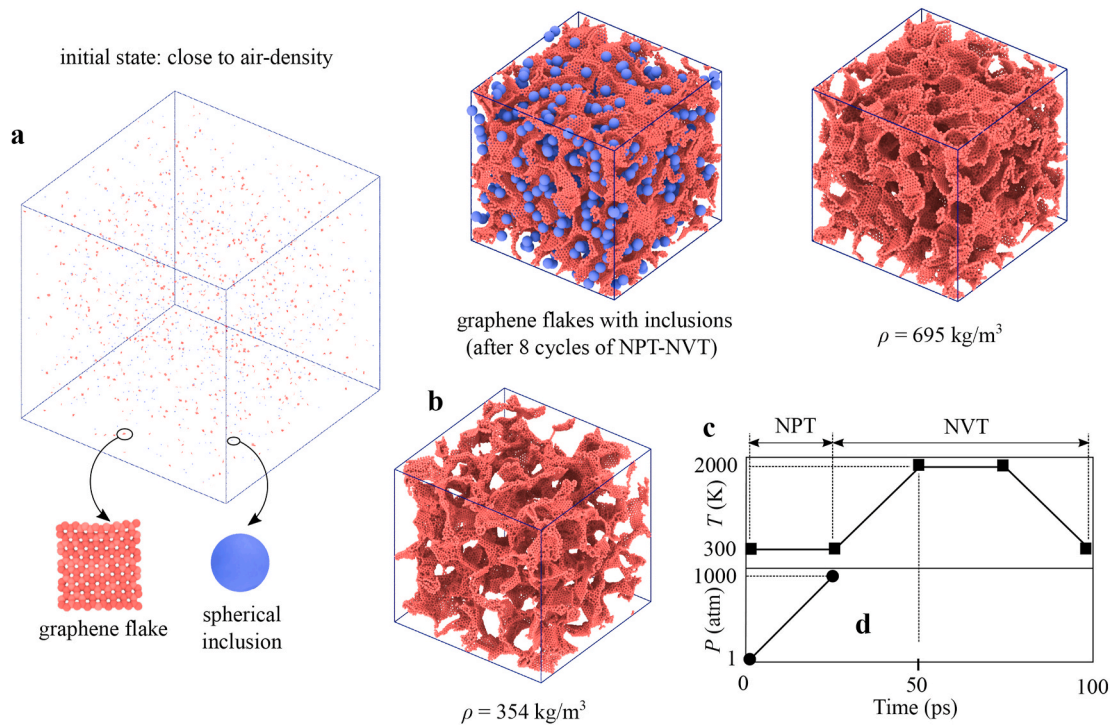


Fig. 20. (a) Schematic representation of construction of the MD simulation model of graphene aerogel of density of 695 kg m⁻³. (b) The GA model of density of 354 kg m⁻³. (c) A typical NPT-NVT cycle. (d) Pressure versus simulation time (Adapted with permission from Ref. [176]. Copyright 2020 American Chemical Society.).

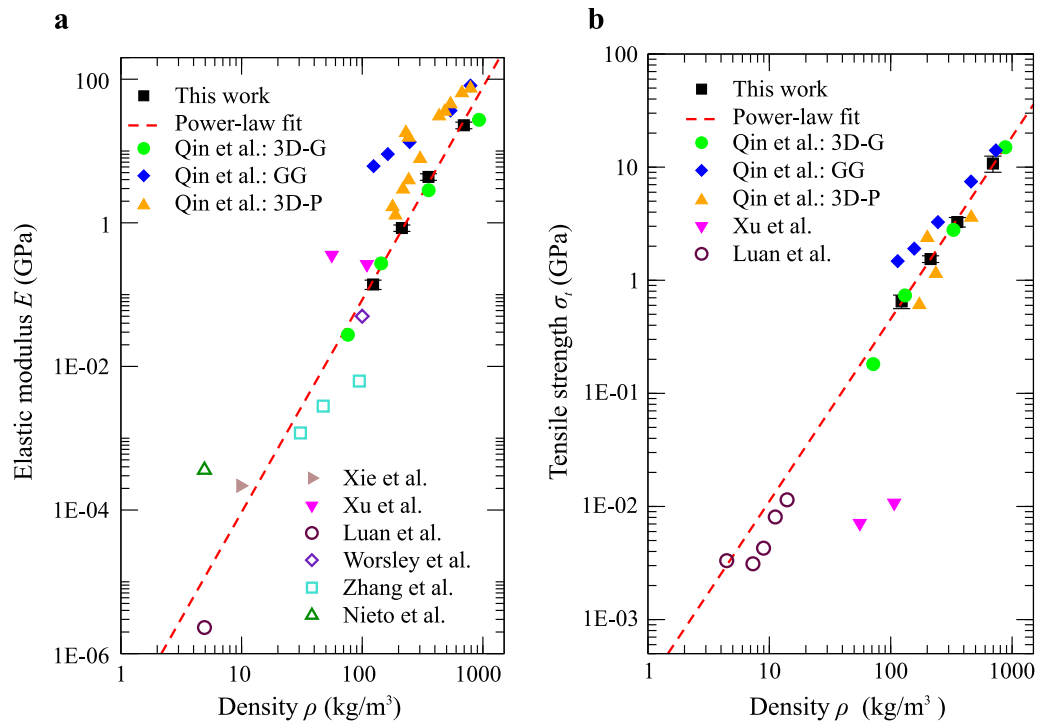


Fig. 21. Power law relationship: (a) elastic modulus E versus density ρ and (b) tensile strength σ_t versus density ρ with standard error bars (Adapted with permission from Ref. [176]. Copyright 2020 American Chemical Society.).

thermal stability and electrical conductivity through effective medium theory (EMT) (see Fig. 23). In this study, EMT is an approach used to circumvent the computational limitations of molecular dynamics. In this approach, the graphene nanosheets have fixed quantitative values that do not change throughout the simulation. However, it does not indicate an accurate measure of graphene distribution within the model.

Although the results are quite higher than those obtained through experiments, the study states that it is possible to attain agreeable results by applying the random Monte Carlo method and accounting for the pore distribution. Further implementations of this suggestion are yet to be carried out.

Meija et al. [183] studied an aerographite tetrapod under various

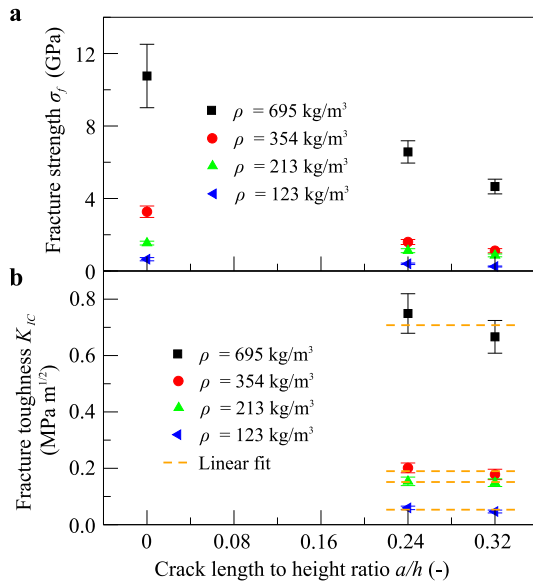


Fig. 22. Influence of the crack depth to height ratio a/h on (a) the fracture strength σ_f and (b) fracture toughness K_{IC} (Adapted with permission from Ref. [176]. Copyright 2020 American Chemical Society.).

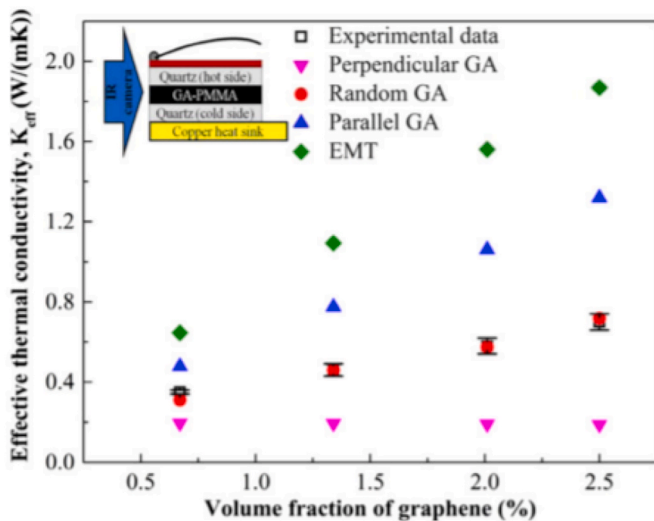


Fig. 23. Effective thermal conductivity as a function of graphene volume fraction (Reprinted with permission from Ref. [182]. Copyright 2015 Elsevier.).

mechanical conditions, both experimentally and computationally, to ascertain their behavior. Initially, the tetrapod was subjected to a bending experiment. According to this work, before the model buckled under the load, the response was governed by the transverse deformations of adjacent arms close to the center, thus subsequently contributing to the central buckling, as shown in Fig. 24. In the next set-up, only the central arm is subjected to a reversible buckling experiment, and the results are compared with the experimental approach. According to the results, it can be seen that the arm can buckle as well, but only if the joint rotational stiffness is sufficiently high. Also, the study shows that the arm eventually recovered from the flexural folding elastically. This property has been well documented by Favo et al. [184] for the multi-walled CNTs, although on a smaller model. The tetrapod was also subjected to tensile and compressive loads, both with fixed and sliding boundary conditions.

Although not extensive, the little available literature provides an insight into the prominent mechanical characteristics of a graphene-

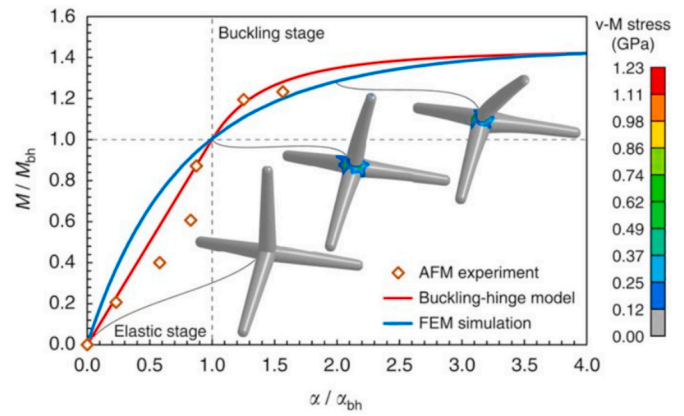


Fig. 24. Buckling results from FEM simulation compared to results from experiments and analytical models [183].

based aerogel and provides direction for their suitable applications. With more research, both computationally and experimentally, it is possible to ascertain more properties, capabilities and potential applications.

The enhanced thermal conductivities of graphene aerogels via supercritical ethanol drying were ranges of 0.0363–0.0667 W/m K, which are lower than polyurethane foam [185]. Silica aerogel has been studied experimentally and numerically for shockwave and high-energy absorption applications. However, the shockwave response of graphene aerogels has not been investigated. The mechanical properties and behavior of graphene aerogels can be studied numerically to explore the high-energy absorption application area. Moreover, graphene aerogel nanocomposites with reinforced materials, such as carbon nanotubes and natural fibers, can be studied using MD simulations, coarse-grained, micromechanical multiscale, and continuum mechanics modeling. Furthermore, one can propose the sandwich structure of graphene aerogel and carbon nanotube aerogels.

5. Cellulose aerogels

Cellulose are a naturally occurring polymers which have the advantages of being biodegradable, biocompatible, non-toxic material [186]. Based on the application, the cellulose have been developed to form aerogels, hydrogels, composites, and nanofibrils amongst the many materials used to enhance their properties. In particular, cellulose-based aerogels are advantageous since they are renewable, biodegradable, highly porous and lightweight. The major applications for cellulose aerogels are in the area of biomedical science as drug carriers, in environmental science as contamination containers, and aeronautical engineering. Apart from this, the cellulose aerogels are also used to produce carbon aerogels.

Since Kistler [4] experimented on the protein strains in the 1930s to produce the first cellulose aerogels, they have seen a significant improvement in their manufacturability and implementation. During the synthesis of cellulose aerogels, instead of the supercritical drying, the freeze-drying technique is preferred. Also, unlike the silica aerogels that are too brittle and the other synthetic polymer-based aerogels which are too toxic for biodegradable purposes, cellulose aerogels are easier to manufacture, biocompatible and have a higher surface area. Although several experimental results have been obtained to observe their properties [186–195], the computational results are very few because of technological constraints and poor mechanical performances.

There have been several computational studies, which have investigated the mechanical properties of cellulose nanofibril/composites individually. But till now, they have not been extrapolated to include the cellulose aerogels. However, the interest in the simulation aspect of these materials has flared in the past decade. To the best of our

knowledge, surprisingly, these materials have never been investigated using MD simulations. Therefore, further exploring the computational modeling and simulations of the cellulose aerogels could lead to various other applications.

5.1. Continuum modeling

In 2015, Baillis et al. [196], constructed a heat transfer model which encompasses the mean free theory and numerical finite analyses to analyze the radiation equivalent conductivity, a factor influencing various properties of the cellulose aerogels. This model takes into account the heat radiation and conduction in both solid and fluid phases of the aerogel. An ideal tetrakaidecahedron solid structure with cylindrical struts is used to demonstrate the solid phase in an open cell and mean-free path of gas molecules to represent the fluid phase. The model also accommodates the Knudsen effect while performing computational iterations.

Rege et al. [197] proposed a micro-mechanical model to simulate cellulose aerogel. To the conscious knowledge, this is the unique model available that has the capability to computationally model, simulate and analyze the mechanical properties of the cellulose aerogels. It utilizes the Euler-Bernoulli beam theory for large deflections to denote the bending in the walls. A network formed of square cells with various pore sizes represents the cellulose aerogels in this model. It is a multi-scale model because it needs the value of nitrogen adsorption isotherm, acquired either through experiments or numerical analysis, to account for the cells' pore size variation. The non-linear bending and subsequent collapse of these cells form the basis of this technique. In this study, both micromechanical simulations and experiments were conducted to add credence to the results. It also considers two different aqueous cellulose constituents and analyzes the specimen/model under compressive stress. The simulation results agree quite well with that of the experimental results, as shown in Fig. 25.

Moreover, Rege et al. [198,199] explained the dependence of damage and failure in cellulose aerogels on the pore size distribution. In this work, the cells start collapsing under the load progressively from the smaller to the larger ones. Hence, before complete failure, some active and damaged cells simultaneously co-exist in the structure. The studies have also shown the behavior and mechanical properties of cellulose aerogels under tensile loading. Under tensile loading, the cells are affected by the dual impact of bending and axial deformations. Once the normal stress reaches a critical value, the fibrils were forming the cell network break apart, inducing failure.

In 2019, Dosta et al. [200] made use of another model that utilizes the bonded-particle method, wherein the material is considered as a set of discrete primary particles connected by solid or liquid bonds. In each iteration, certain breakage criteria are observed, and once even a single

of these criteria are met, the bond is destroyed and discarded from further computations. For quasi-brittle material simulations, Ma et al. [201] demonstrated a displacement softening contact model. Although in this study, the Alginate aerogels are simulated after forming a new functional model to represent the bonds. The model with various concentrations is tested under uniaxial compression loads and validated by experimental results. The only difference between the experimental and simulated results is that the recovery strain remains the same for all concentrations in the simulated output, as shown in Fig. 26. The model is also observed under three-point bending tests, both experimentally and numerically. As can be noted, the simulated results underestimate the force but predict the material behavior correctly.

The thermal conductivity of cellulose aerogels is ranging between 0.018 and 0.075 W/m K, which is lower than polyurethane foam [202]. Cellulose aerogels nanocomposites with reinforced materials, such as carbon nanotubes and natural fibers, can be studied using MD simulations, coarse-grained, micromechanical multiscale, and continuum mechanics modeling. Moreover, one can propose the sandwich structure of cellulose aerogel with graphene or carbon nanotube aerogels.

Since the biopolymers are more eco-friendly, non-toxic to manufacturers and have varied applications, there has been an increase in the interest to develop them of late. Although the nature of these materials and their behavior itself are a constraint to the computational studies, there have been good efforts and strides made in the research as exemplified above. But further numerical and experimental studies are required to develop and understand the material in the particular biomedical application.

6. Conclusions

The relatively new classification of the ultralight materials is called aerogel. In the present work, the obtained mechanical and thermal properties of aerogels using the mechanical modeling and simulations are thoroughly reviewed. Moreover, their applicability for specific goals has been critically discussed. The silica aerogels are the best-developed aerogels in the computational studies since they widely used in many applications, wherein the understanding of how and why they fail has been an extensively researched subject. Several approaches to the mathematical modeling of silica aerogels, such as molecular dynamics, coarse-grained and continuum mechanics modeling, have been discussed in depth. These approaches are significantly useful in understanding aerogels' deformation behavior and evaluate physical parameters at different length scales. Furthermore, the above techniques can be effectively used to reinforce the silica aerogel for specific applications.

Carbon and graphene aerogels are the computationally less developed since they were discovered in the late 20th century and early 21st

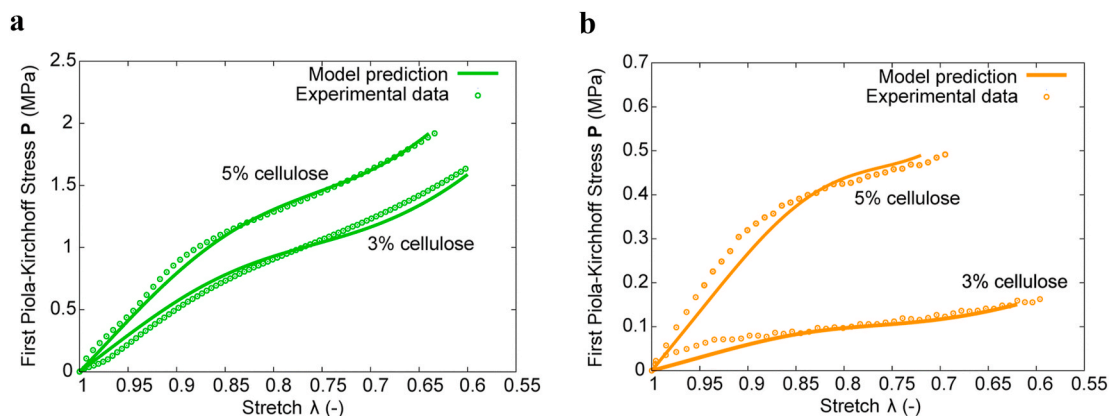


Fig. 25. Model predictions of the constitutive response of cellulose aerogels, prepared from (a) ZC-route and (b) CT-route compared with experimental data (Reprinted with permission from Ref. [197]. Copyright 2016 Royal Society of Chemistry.).

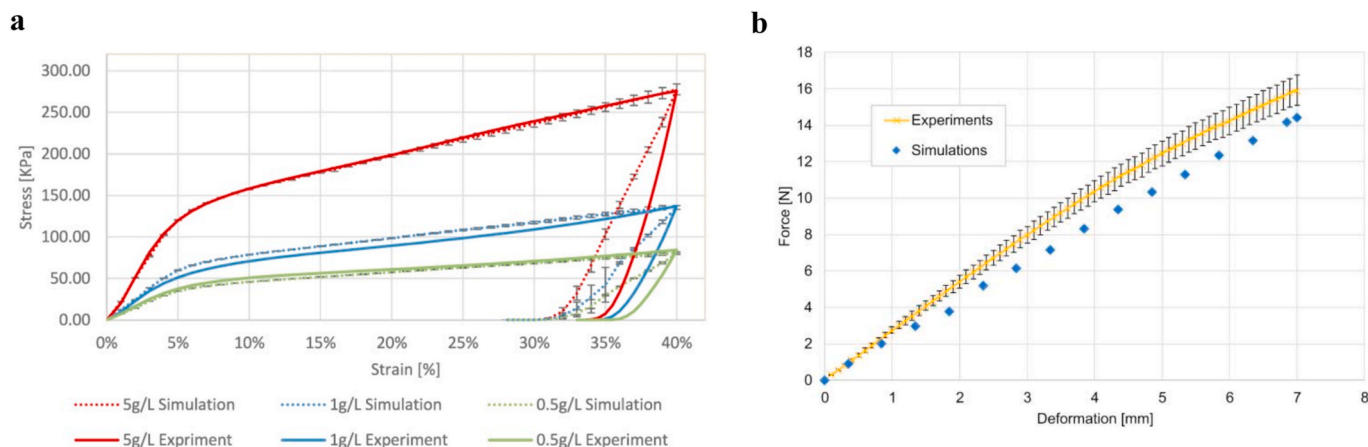


Fig. 26. (a) Compressive stress-strain comparative study at different cellulose concentration and (b) three point bending test deformation results [200].

century, respectively. However, research efforts have already been made to understand the materials experimentally, and also simulation results show the advantages of reinforcing these materials with nanotubes and oxides. Cellulose aerogels or the biopolymer aerogels are the least developed among the four categories of aerogels in the literature. This is because of the unpredictable behavior of organic cellulose. Recently, research shows that cellulose independently has been studied exclusively to get a better understanding. But with the large population becoming eco-conscious, there has been a flare in the research interest in the cellulose aerogels as well. Their applicability as oil adsorbers to clean the oil spills in water has been researched extensively. But with further computational efforts, there is a possibility to explore the extent of the unpredictability of these materials.

To the best of our knowledge, surprisingly, carbon and cellulose aerogels have never been investigated using MD simulations. Therefore, further exploring the computational modeling and simulations of the various aerogels and their composites pave the way toward numerous unexplored applications.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study. However, the data used to support the findings of this study may be made available from the corresponding author upon request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the computing time granted by the JARA Vergabegremium and provided on the JARA Partition part of the supercomputer CLAIX at RWTH Aachen University. The authors, V. G. Parale and H.-H. Park would like to thank that this work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2020R1A5A1019131).

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